

Homework 2

MAE 384

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Q1

Problem Statement: I am to fit an equation that runs through all the points of the given data through various methods.

```
%% Q1

%Givens
reynolds = [2 20 200 2000 20000 40000 200000 400000 2000000];
drag = [13.9 2.72 0.8 0.401 0.433 0.47 0.4 0.0775 0.214];

%Polyfit
x = log(reynolds);
y = log(drag);
coef = polyfit(x,y,1);
xx = linspace(min(x),max(x),11);
yy = polyval(coef,xx);
plot(x,y,'*',xx,yy)

%Vandermonde
xxv = linspace(min(x),max(x));
v = vander(x);
ydiv = y';
a = v\ydiv;
yyv = polyval(a,xxv);
hold on
plot(x,y,'*',xxv,yyv,'b'); grid on

%Spline
spl = spline(x,y,xxv);
plot(x,y,'*',xxv,spl,'g'); grid on

%Quadratic Fit
coef4 = polyfit(x,y,4);
xx4 = linspace(min(x),max(x),11);
yy4 = polyval(coef4,xx);
plot(xx4,yy4,'yellow')

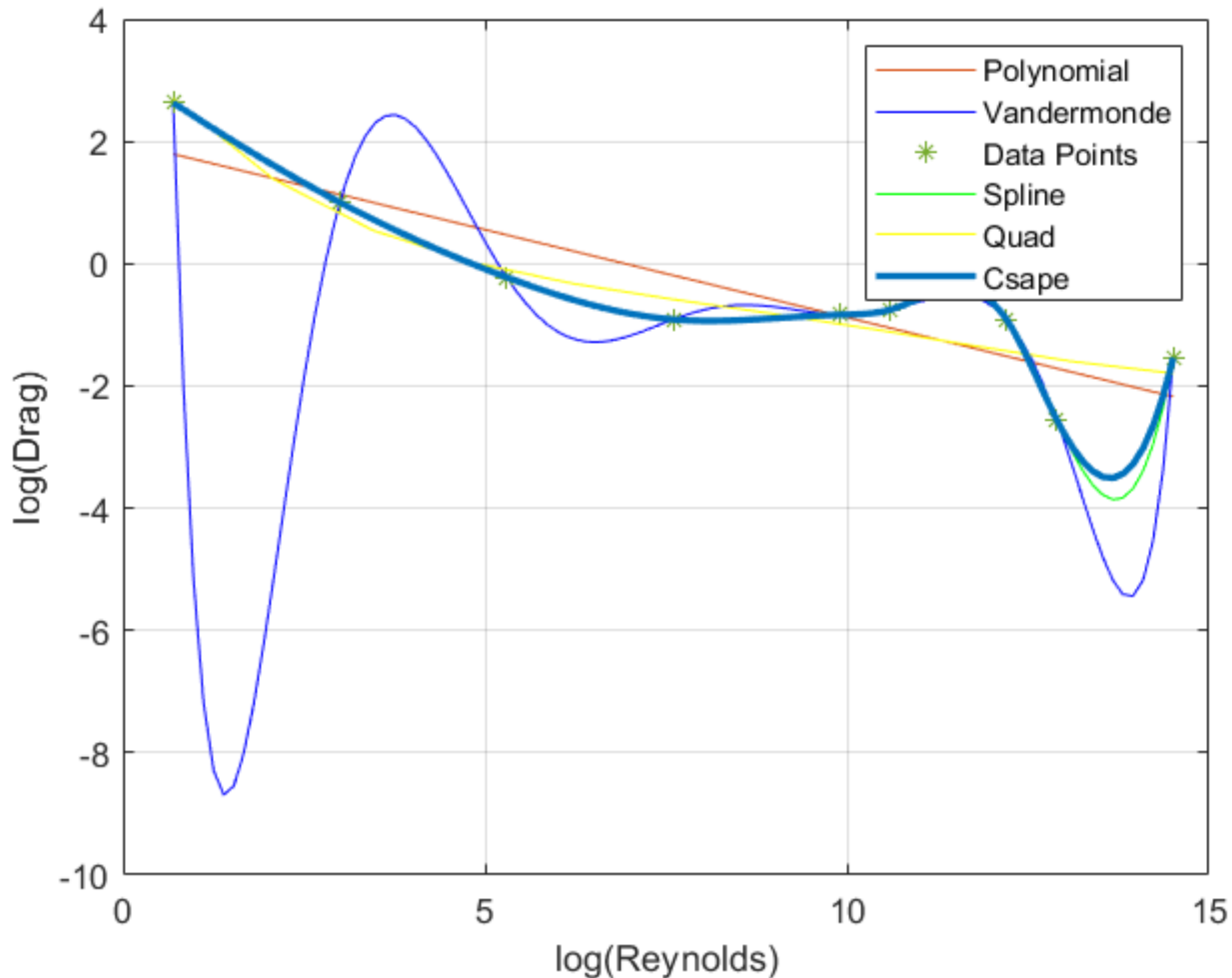
%CSape
pp = csape(x,y);
fnplt(pp)

xlabel('log(Reynolds)')
```

```

ylabel('log(Drag)')
legend('','Polynomial','','Vandermonde','Data Points','Spline','Quad')
hold off

```



Solution: For the polynomial and quad fits, I used polyfit with the order of the equation I want to go through the data points. For the vandermonde fit, I created a vandermonde matrix of the data and then left divided it with the derivative of the y vector to get the equation. For the csape line, I used the csape function and then fnplt to put it onto the plot.

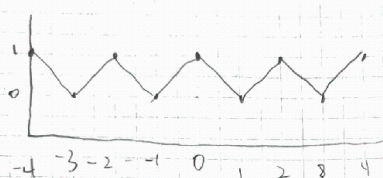
Q2

Problem Statement: I am to create a fourier analysis for a periodic triangle wave.

Solution: On paper I determined the coeffs, a0 and ak1 using the given formulas. I then inputted those into Matlab, and created a for loop with the fourier formula to run it through the given 1,3, and 21 times.

2.

$$f(t) = \begin{cases} 1+t & -1 \leq t \leq 0 \\ 1-t & 0 \leq t \leq 1 \end{cases}$$



A) Find Fourier coefficients for the function

Formulas for continuous Fourier series

$$f(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(k\omega t) + b_k \sin(k\omega t))$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$\omega = \frac{2\pi}{T} = \pi$$

$$a_k = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(k\omega t) dt$$

$$b_k = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(k\omega t) dt = 0 \text{ as even function}$$

$T=2$

$$a_0 = \frac{1}{2} \int_{-1}^1 (1+t) dt = 1/2$$

$$= \frac{1}{2} \int_{-1}^1 (1-t) dt = 1/2 = a_0$$

$b_k = 0$ since even

$$A_k = \frac{2}{2} \left(\int_{-1}^0 (1+t) \cos(k\omega t) dt + \int_0^1 (1-t) \cos(k\omega t) dt \right)$$

$$= \left(\frac{1}{k\omega} \sin(k\omega t) + \frac{t}{k\omega} \sin(k\omega t) + \frac{\cos(k\omega t)}{\omega^2 k^2} \right) \Big|_{-1}^0$$

$$- \left(\frac{1}{k\omega} \sin(k\omega t) + \frac{t}{k\omega} \sin(k\omega t) - \frac{\cos(k\omega t)}{\omega^2 k^2} \right) \Big|_0^1$$

$$\frac{\cos(0)}{\omega^2 k^2} - \frac{\cos(-k\omega)}{\omega^2 k^2} - \frac{\cos(k\omega)}{\omega^2 k^2} + \frac{\cos(0)}{\omega^2 k^2}$$

$$= \frac{2}{\omega^2 k^2} - \frac{2 \cos(k\omega)}{\omega^2 k^2} = \boxed{\frac{2 - 2 \cos(k\omega)}{\omega^2 k^2} = a_k}$$

%% Q2

%Original Function

```
t1 = linspace(-1,0,1000);
t2 = linspace(0,1,1000);
f1a = 1+t1;
f2a = 1-t2;
hold on
plot(t1,f1a,t2,f2a);
```

%Fourier Series

```
t = linspace(-2,2,1000);
ks1 = 1:1;
ks2 = 1:3;
ks3 = 1:21;
```

%Coeff

```
a0 = 1/2;
ak1 = (2-2*cos(ks1.*(pi)))/((pi^2)*(ks1.^2));
ak2 = (2-2*cos(ks2.*(pi)))/((pi^2)*(ks2.^2));
```

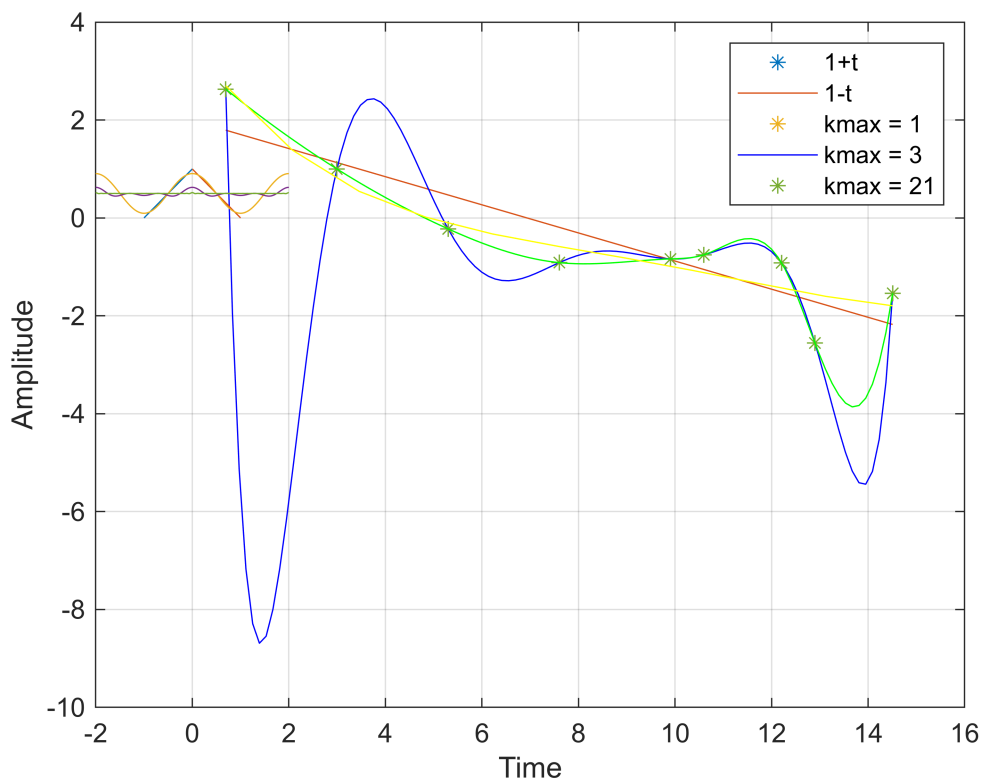
```

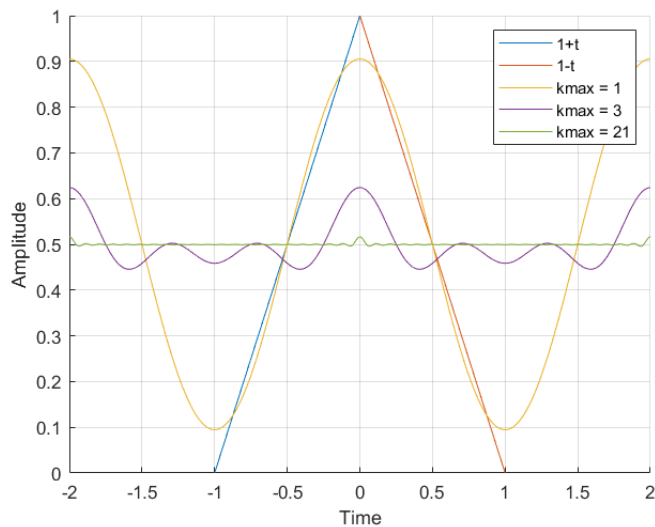
ak3 = (2-2*cos(ks3.*(pi)))/((pi^2)*(ks3.^2));

for i = 1:1000
    f1(i) = a0 + sum(ak1*cos(pi.*ks1*t(i)));
    f2(i) = a0 + sum(ak2*cos(pi.*ks2*t(i)));
    f3(i) = a0 + sum(ak3*cos(pi.*ks3*t(i)));
end

%Plot fourier approxs on top of orig function
plot(t,f1,t,f2,t,f3);
grid on
xlabel('Time')
ylabel('Amplitude');
legend('1+t', '1-t', 'kmax = 1', 'kmax = 3', 'kmax = 21')

```





Solution: I have no idea what went wrong here. Kmax1 looks good but each subsequent one gets worse and I have no idea why. I'm pretty sure I did everything right but I messed up somewhere and I have 0 idea where.

From my code, 0.00001 would be best but I know that's wrong so I can't make an educated guess.

Q3

Problem Statement: I am to interpolate data with the given formula to get an estimate for the mass of the earth.

Solution: I integrated the formula to get an equation. I used the trapz function to get an answer. For Part B I divided the data into 50 equally sized segments and then repeated the trapz method and then created a function for simpson's method and did it for the new divided data.

```
%% Q3

%Givens
r = [0 800 1200 1400 2000 3000 3400 3600 4000 5000 5500 6370];
p = [13000 12900 12700 12050 11650 10600 9900 5500 5300 4750 4500 3300];

rm = r*1000; %Convert into meters so we can use rho which is in meters

%Bounds
a = 0;
b = 6370*1000;
n = 11;
%Part A
%Trapezoid Rule
h = (b - a)/n;
XtrapzA = a:h:b;
functA = ((p).*(4)*(pi).*(rm.^3))/3;
Atrapz = trapz(XtrapzA,functA);
fprintf('The mass of the earth found through trapz in Part A is %e kg \n', Atrapz);
```

The mass of the earth found through trapz in Part A is 7.749727e+30 kg

%Part B

```
fxB = linspace(0,6370,50);  
pB = linspace(13000,3300,50);  
XpartB = 1000*(fxB);  
functB = ((pB).*(4)*(pi).*(fxB.^3))/3;  
s = spline(rm,p,XpartB);  
Btrapz = trapz(XpartB,functB);  
fprintf('The mass of the earth found through trapz in Part B is %e kg \n', Btrapz);
```

The mass of the earth found through trapz in Part B is 9.034829e+21 kg

```
fs = @(x) (max(pB)*4)*(pi*((x)^3))/3;  
hs = (b-a)/2;  
simpson = (hs/3)*(fs(a)+(4*fs((a+b)/2)+fs(b)));  
fprintf('The mass of the earth found through simpsons rule in Part B is %e kg \n', simpson);
```

The mass of the earth found through simpsons rule in Part B is 2.241453e+31 kg

Discussion: I know this is wrong, I have 0 idea why this is wrong but I know it is wrong. The data is too far off even though I'm pretty sure I did everything right from notes and recitations. I just flat out do not know what I did wrong, again. There is my discussion, take it or leave it.